Example 8 Find the cost of fencing a square plot of side 12 m at the rate of ₹15 per metre.

Solution

Side of a square plot = 12 m

Perimeter of the square plot = (4 × s) $units$

= 4 × 12 = 48 m

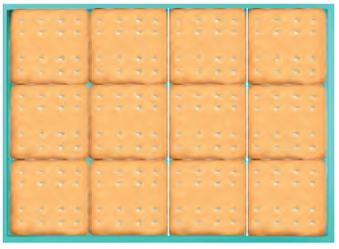
Cost of fencing the plot at the rate of ₹15 per metre = 48 × 15 = ₹720

Try these

i) Find the breadth of the rectangle with perimeter 14 m and length 4 m.

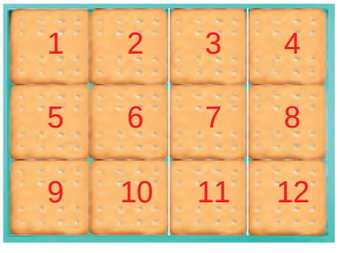
ii) The perimeter of an isoseles triangle is 21 cm. Find the measure of equal sides given that the third side is 5 cm.

3.3 Area



Recall the 'Apoorva and her biscuits arrangement' in the beginning of this chapter. We do not know the measure of the side of the biscuit. But we know it is in the shape of a square. Let its side be 1 unit. The tray can hold 12 square biscuits (square units). That is, 12 square biscuits (square units) occupy the entire surface of the tray. This space of the tray is called the ***Area*** of the tray. Thus, the area of any closed shape is the surface occupied by the number of unit squares within its boundary. Suppose each side of a biscuit is of 1 inch length, then the area of the tray is 12 square inches.

3.3.1 Area of a Rectangle



As the above tray is in the shape of a rectangle, split it into small and uniform squares called unit squares. There are 4 unit squares along its length and 3 unit squares along its breadth. Totally, 12 square biscuits occupy this rectangle and so the area of this rectangle is 12 square units.



Suresh brought a groundnut burfi packet to school as snacks to eat during the break. As he peeled off the cover, he saw that there are 3 rows and each row has 5 square pieces. What he sees here is that the total number of small squares in the given burfi is 15. So, the area of the given rectangular burfi is 15 square burfi pieces.



Tamizhazhagi wants to share a chocolate bar with her friends on her birthday. The chocolate bar that she has bought has 5 square pieces horizontally and 4 square pieces vertically. She finds that there are 20 identical unit square chocolate pieces so that she can give it to 19 of her friends and have 1 for herself.

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